

Piyush Deshpande

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EDUCATION

Master of Science, Purdue University, West Lafayette
Computer and Information Technology (GPA: 3.88/4)

August 2024 - Dec 2026

Bachelor of Technology, Vishwakarma Institute of Technology, Pune
Electronics and Telecommunications (CGPA: 8.98/10)

August 2019 - May 2023

SKILLS

Programming Languages: C++, SQL, Python

Frameworks & Libraries: TensorFlow, Scikit-learn, OpenCV

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Power BI

Relevant Coursework: Object-Oriented Programming, DBMS, Machine Learning Algorithms, Agile Fundamentals

Areas of Interest: Data Analysis, Data Science, Machine Learning (ML), Artificial Intelligence (AI)

Soft Skills: Analytical Skills, Result-driven leadership, Strong Communication, Problem Solving, Adaptability

EXPERIENCE

Software Engineer

Purdue University, Department of Forestry and Natural Resources

March 2025 - Present

West Lafayette, IN

• Forest Inventory Data Pipeline & Warehousing for Invasive Pest Detection:

- Engineered a Python-based data pipeline for invasive pest detection, programmatically extracting 80+ granular forest inventory attributes (e.g., volume, growth, mortality) for specific pest-host species across all US states via the FIA API.
- Developed a sophisticated data merging algorithm to resolve overlapping evaluation periods, ensuring de-duplicated, current, and historical forest health data (back to 2006) for timely analysis of pest impacts. This involved processing files sorted by evaluation code for "latest-wins" accuracy.
- Designed and populated a Microsoft SQL Server data warehouse using a star schema (FIAMeasurements and dimensions like Location, MeasurementType, Species). This robust ETL process (utilizing pandas and pyodbc) provides critical, accessible insights for monitoring vulnerable tree populations.

• LiDAR SLAM Application Optimization:

- Reduced memory consumption by 40% for a large-scale LiDAR SLAM application, enabling more efficient batch processing of large point cloud datasets.
- Achieved this by architecting and implementing incremental downsampling and scan purging strategies in the C++ (15k+ LOC) codebase.
- Systematically debugged and resolved critical C++ library linkage issues (Boost, PDAL, PCL) by manually configuring CMake build flags.

Analyst

Coforge

July 2023 - July 2024

Pune, India

- Worked closely with cross-functional teams, automating approval workflows at bp, transforming manual processes into an optimized Python-based system. Wrote SQL scripts for data validation, cutting approval time by 40% and billing errors by 25%.
- Optimized data pipelines with Python and SQL, boosting database efficiency by 30% and reducing query runtime by 15%.

Software Engineer Intern

IdeaS - A SAS Company

July 2022 - December 2022

Pune, India

- Supported development of Spring Boot (MVC-based) revenue optimization platform; implemented backend REST APIs and MySQL test databases, enhancing test coverage and system reliability by 25%.
- Demonstrated strong verbal and written communication skills through documentation and Postman-based API validations, leading to smoother client handovers and accurate QA.

PROJECTS

Computer Vision-based Staircase Detection for Safe Mobility of Visually Impaired People

Engineered a smart cane system using Python and OpenCV to help visually impaired individuals detect staircases. Designed a feature extraction pipeline using SIFT and PCA, reducing processing time by 30% and boosting accuracy to 95%. Demonstrated strong problem-solving and communication skills by documenting edge cases and improving system robustness. ([Link](#))

An Electromyography-based system for Sitting and Walking Postures

Implemented a signal classification model to analyze Electromyography (EMG) signals for sitting/walking postures. Preprocessed EMG signals using SciPy and NumPy, extracted features, and evaluated Decision Trees, SVM, KNN, and Random Forest classifiers. Achieved 99.19% accuracy using the Random Forest algorithm. ([Link](#))